pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Metadata Filters and Many Documents

WEEK 2 · DAY 4

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Today: a corpus of four

Four documents, one corpus

Corpus	3,851 chunks at 512/64
Documents	a 10-K and three indentures
Questions	25 development questions
Answers	8 saved answers, replayed
Held-out	not opened until day 11

Today

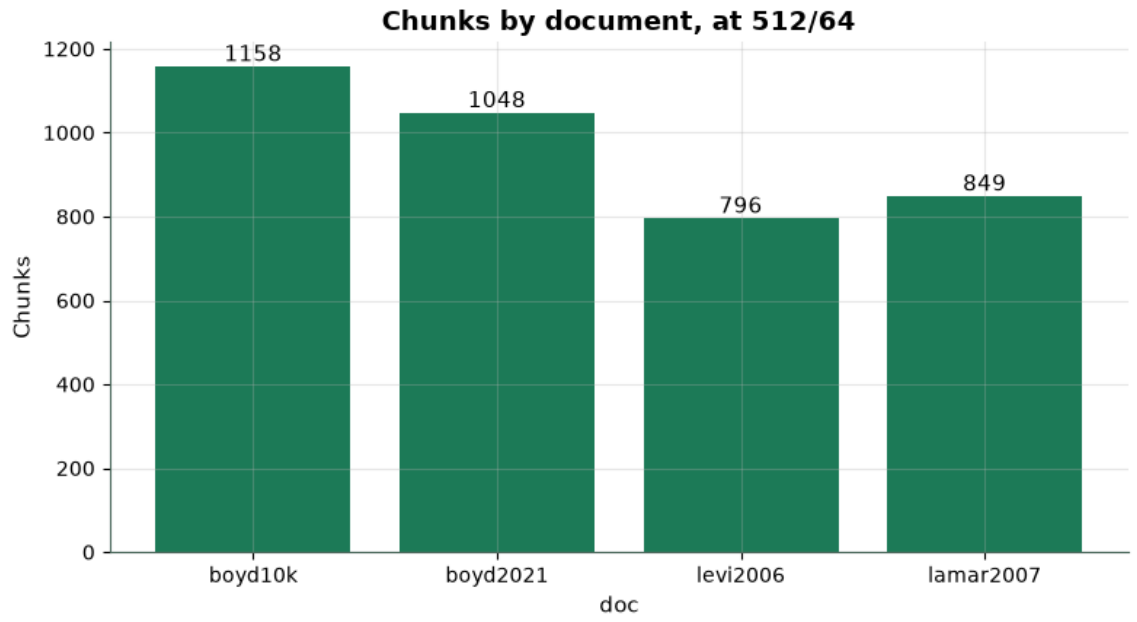
1. One corpus, four documents
2. Metadata as columns, filters as masks
3. A question with no filter
4. Filter before search
5. The join the corpus allows
6. A question the corpus cannot answer
7. The wrong document

Say which document a question is about with a filter, and read the document in the citation.

How a corpus with metadata works

The problem	Text from the wrong document can match a question as well as the right one, and nothing in a chunk's words says which document it came from
The method	Cut each document inside its own units (items, sections), stack the tables, and give every row its document, unit, page, and as-of date
You control	The documents, the unit each is cut inside, and each document's as-of date
What comes out	One chunk table with the metadata as columns, which every filter reads, and which the committed embeddings match row for row

One corpus, four documents



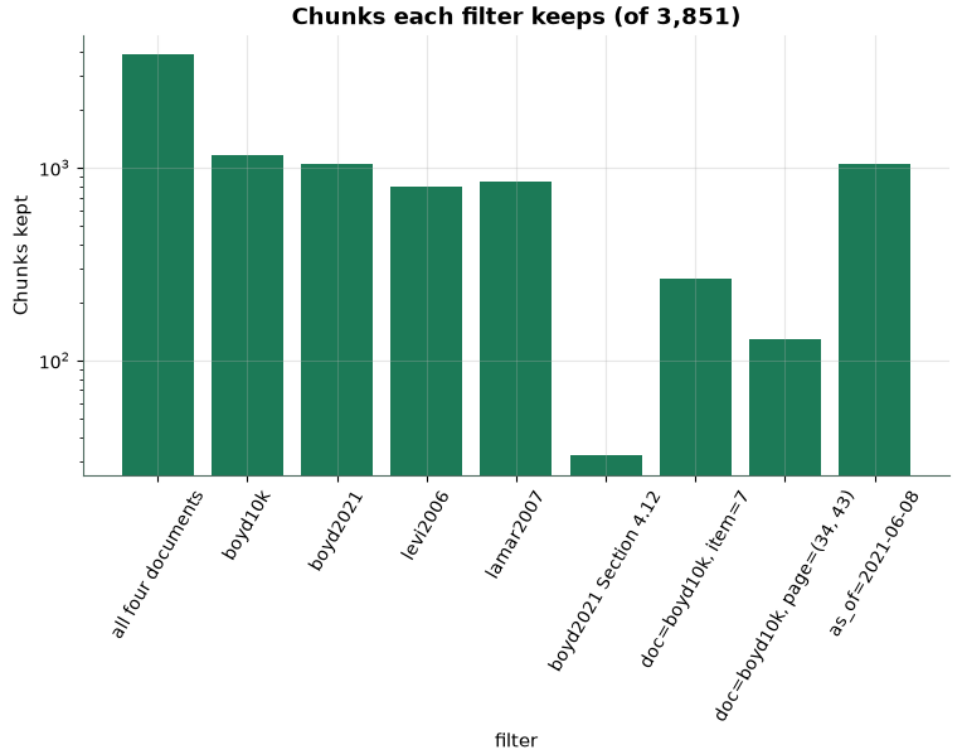
document	cut inside	cuts	chunks	as of
boyd10k	items	24	1158	2025-12-31
boyd2021	sections	108	1048	2021-06-08
levi2006	sections	86	796	2006-03-17
lamar2007	sections	123	849	2007-10-11

A 10-K is cut inside its items, an indenture inside its sections. The last section of each indenture runs on into its exhibits.

How a filter before the search works

The problem	The caller has to say which rows a search may read, and the saying has to act before the search, not after it
The method	filter_chunks reads the metadata and returns a row mask; the retriever is built over the kept rows, with a BM25 index on those rows alone
Not after	A filter applied after the top k throws rows away and returns fewer than k, or none
You control	The fields and the values; several fields together must all hold
What comes out	A mask, then a retriever that behaves as if the corpus held only the chosen rows

Metadata as columns, a filter as a mask



filter_chunks(chunks, **metadata)

```
doc

item  (section)

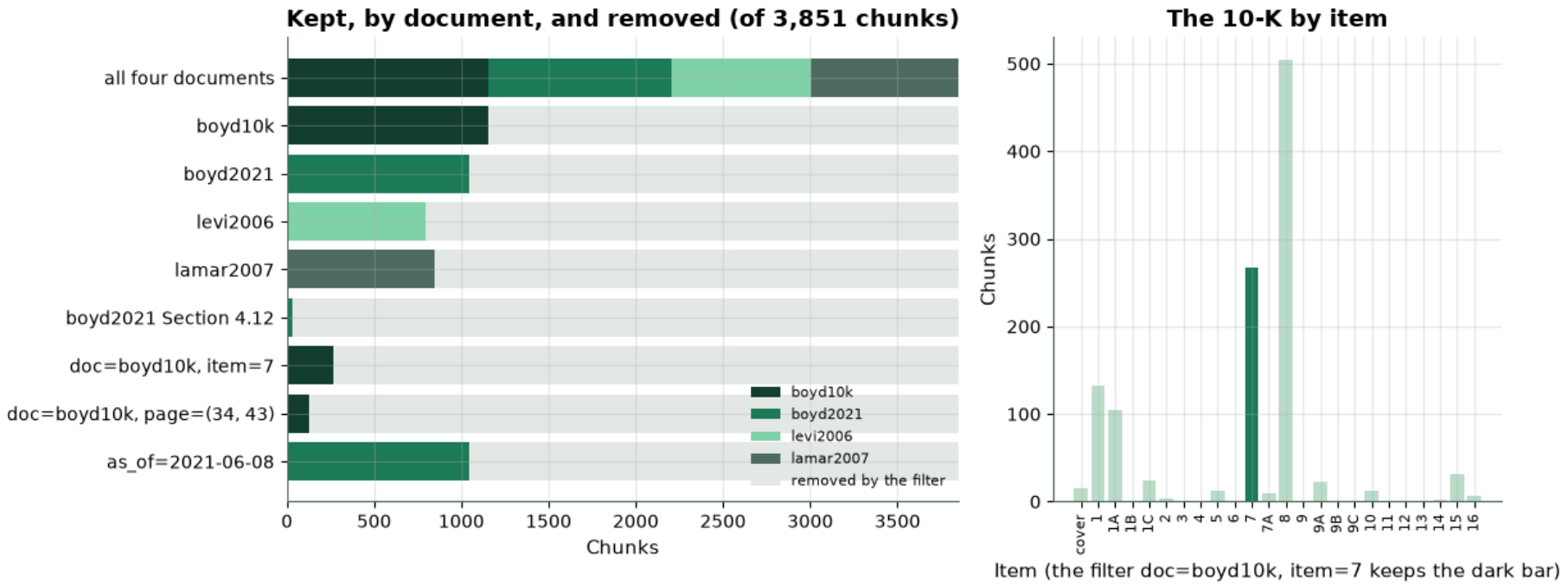
page  (one, or first and last)

as_of  (a value or a list)
```

Returns one True or False per chunk. It searches nothing. The search is handed the rows the mask keeps. An unknown field raises ValueError.

The scopes differ by a factor of 100: 3,851 chunks for all four documents, 32 for one section.

The filter drawn: what it removes before the search




```
Removed before the search: all four documents 0; boyd10k 2,693; boyd2021 2,803; levi2006 3,055; lamar2007 3,002
boyd2021 Section 4.12 3,819; doc=boyd10k, item=7 3,584; doc=boyd10k, page=(34, 43) 3,723; as_of (boyd2021) 2,803
```

A search over a filtered scope never scores a grey chunk: the wrong document cannot win a slot it was never in.

What is the interest rate on the senior notes?

The answer in the saved run

The interest rate on the 6 5/8% Senior Subordinated Notes Due 2015-Series C is 6 5/8% per annum [lamar2007; Section 12.16; c0769] and [lamar2007; Section 12.16; c0700]. The interest rate on the 4.750% Senior Notes due June 2031 is 4.750% per annum [boyd10k; Item 8; p. 77; c0844].

The three chunks behind it

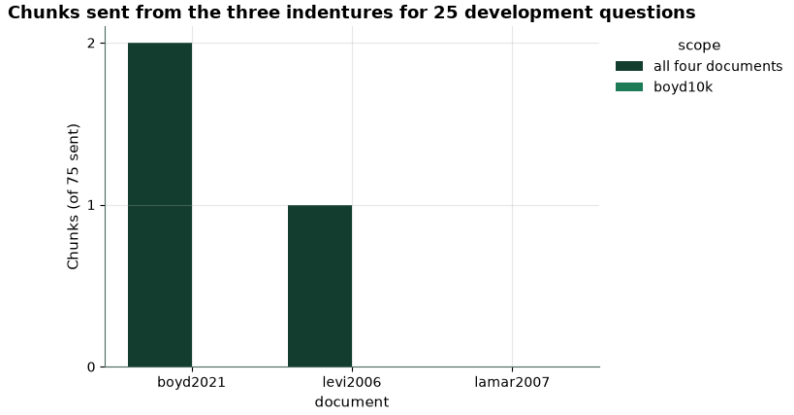
citation	score
[lamar2007; Section 12.16; c0769]	7.21
[lamar2007; Section 12.16; c0700]	7.19
[boyd10k; Item 8; p. 77; c0844]	6.43

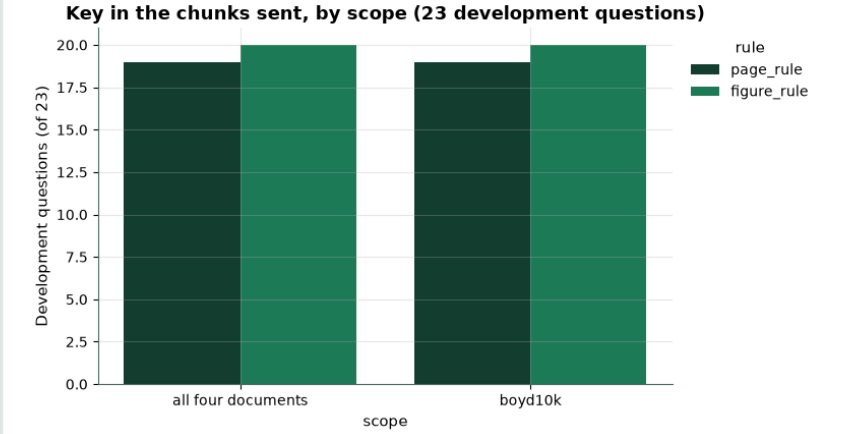
Documents among the chunks: lamar2007, boyd10k; cited: lamar2007, boyd10k. The question named no issuer.

How a filtered retriever works

The order	The mask is made first, from the metadata; then the retriever is built over the kept rows
The method	Their embeddings for the dense list; a BM25 index built on those rows for the keyword list; the fusion and the reranker as on days 6 and 7
The index	BM25 weighs a word by how rare it is among the chunks indexed: four documents give the corpus's weights, the 10-K's rows give the 10-K's
You control	The scope (the mask) and the setting (day 8's carried-forward rerank)
What comes out	k chunks with scores and citations, every one of them from the rows the mask kept

Same question, two filters





With a filter

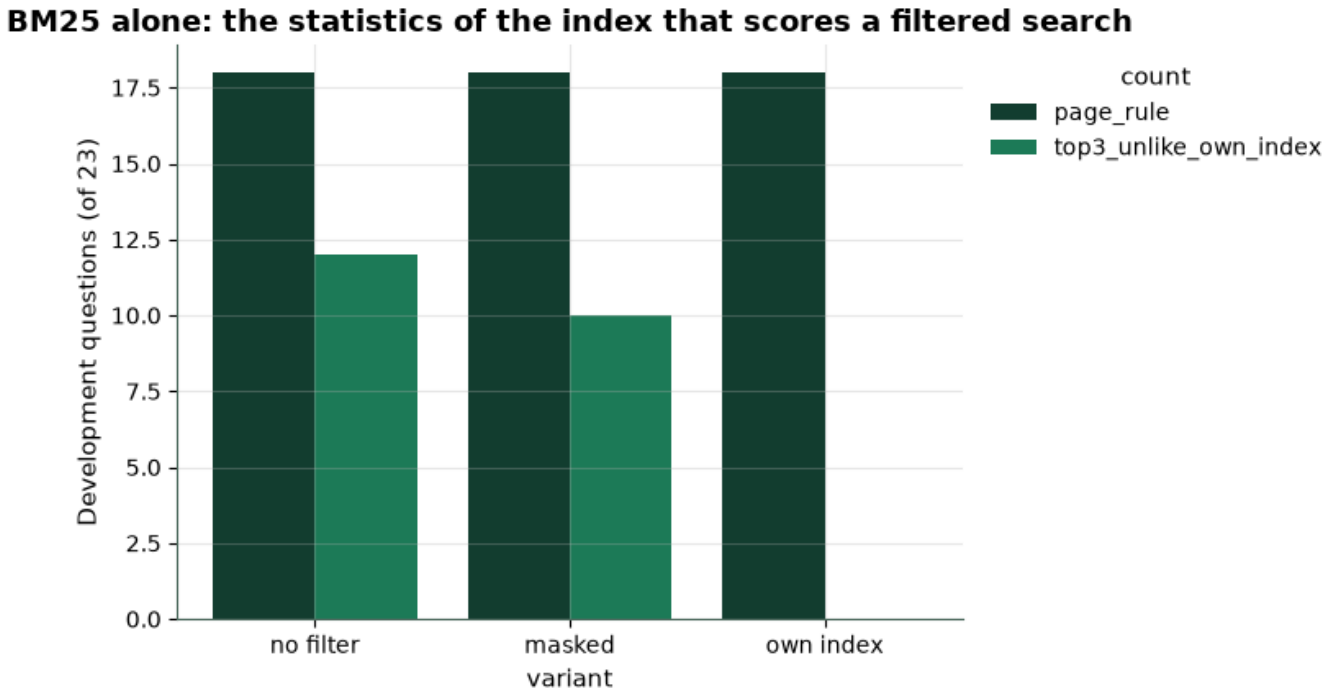
10-K only: cites boyd10k, 4.750%

Boyd indenture only: cites boyd2021, 4.750%

```
page rule / figure rule: all four documents 19/20; boyd10k 19/20
chunks from other documents: all four documents 3; boyd10k 0; in dev08, dev09, dev10
```

Identical chunks with and without the filter: 22 of 25. A filter changes the result for the questions the unwanted documents can answer.

The index follows the rows



Three ways to score a filtered search

variant	page	other	unlike
no filter	18	2	12
masked	18	0	10
own index	18	0	0

Key page found in the top 3: no filter 18; masked 18; own index 18. Top 3 differ between masked and own index for 10 of 23 questions.

Build the index on the rows the search reads. It costs a fifth of a second.

Two documents, two filtered questions

Each answer cited to its own document

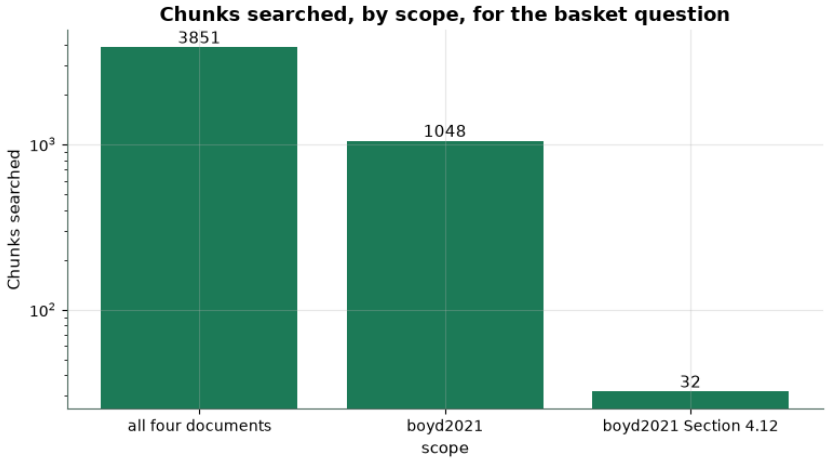
10-K: The 4.750% Senior Notes due 2031 were issued on June 8, 2021, in the aggregate principal amount of $900.0 million [boyd10k; Item 7; p. 39; c0441], [boyd10k; ...

Boyd indenture, Section 4.12: not to exceed the greater of (A) $430,000,000 and (B) 7.5% of the Company's Consolidated Total Assets, calculated on a Pro Forma Basis [boyd2021; Section 4.12; c0549].

```
boyd10k                 key $900.0 million      match

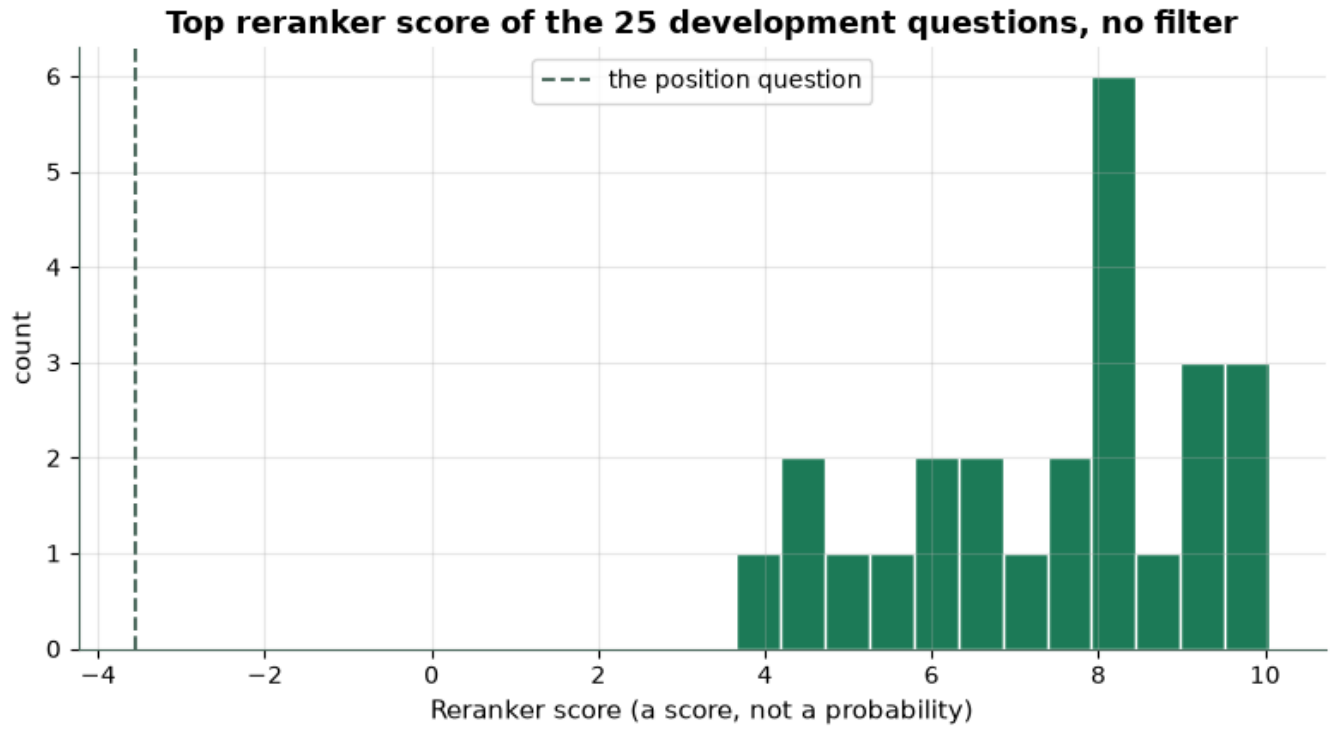
boyd2021 Section 4.12   key $430,000,000; 7.5%  match
```

Chunk with the figures among the three sent: all four documents: not in the three; boyd2021: not in the three; boyd2021 Section 4.12: 3



Figures two public documents state, set side by side. Nothing is said about which is larger.

A position the corpus does not hold



Reranker scores

The position question's three chunks: -3.56, -3.77, -3.79

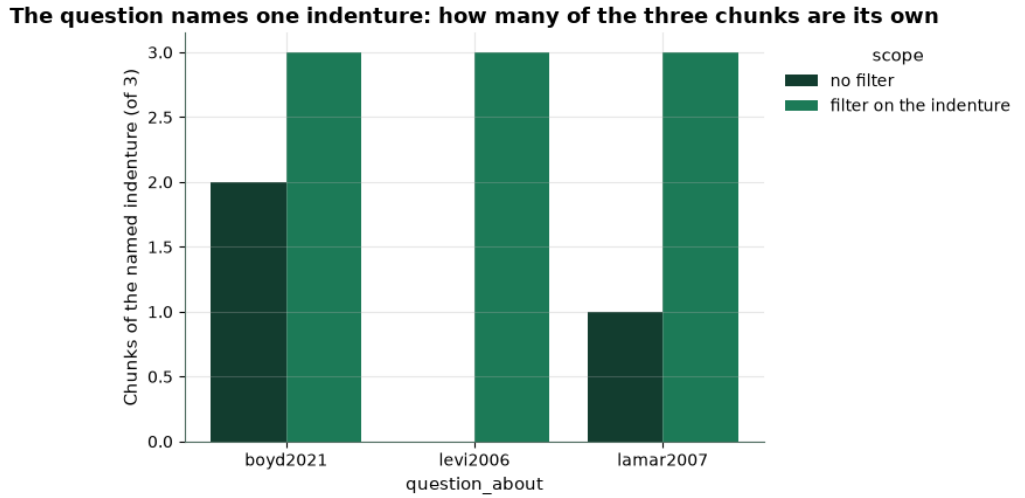
Answerable questions' top score: 3.66 to 10.05.

Questions the 10-K cannot answer: dev24 7.21; dev25 8.10

Answer: NOT IN CORPUS

A score says how well a chunk matches, not whether the question can be answered from it. It is not a probability.

A correct figure from the wrong document



The Levi Strauss question

No filter: "not less than 30% in aggregate principal amount of the Notes then Outstanding" [boyd2021; Section 6.02; c0660].

Filter on levi2006: "at least 25% in aggregate principal amount of the Securities then outstanding" [levi2006; Section 10.13; c0770].

```
no filter           cites boyd2021   asked-about document cited: no  grade: no match
filter on levi2006  cites levi2006   asked-about document cited: yes grade: match
```

Chunks of the named indenture among the three: no filter boyd2021 2; levi2006 0; lamar2007 1; filter boyd2021 3; levi2006 3; lamar2007 3. The citation names the document; a key may not.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=COUNTIFS(chunks[B],"boyd2021")`	chunks of boyd2021: 1,048, equal to Python for all four
`=COUNTIFS(chunks[B],"boyd2021",chunks[C],"4.12")`	a document and a section: 32
`=ROWS(FILTER(chunks[A],chunks[B]="lamar2007"))`	rows kept for lamar2007: 849
`=TAKE(SORTBY(FILTER(...),FILTER(...),-1),3)`	filter first: 3 ids, the same as Python
`=FILTER(TAKE(SORTBY(...),3),TAKE(...)="lamar2007")`	top 3 first, then filter: 1 row

A section such as 4.10 pasted into a General cell becomes the number 4.1: MATCH("4.10",...) returns not found. Paste as Text.

Git: corpus_manifest.json, then the 10-K's rows: changed documents, chunks, matrix_sha256; 47 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the function filters after the search, so it returns fewer than k chunks.

Code that runs is not the same as code that is right.

What you can do now

- A corpus: each document cut inside its own units
- A filter is a row mask, applied before the search; draw what it removes
- Build the index on the rows the search reads
- Read the document in the citation, not only the figure
- A score is not a refusal rule; NOT IN CORPUS is the model's job

Week 2, Day 5 Case study: prompts that must cite